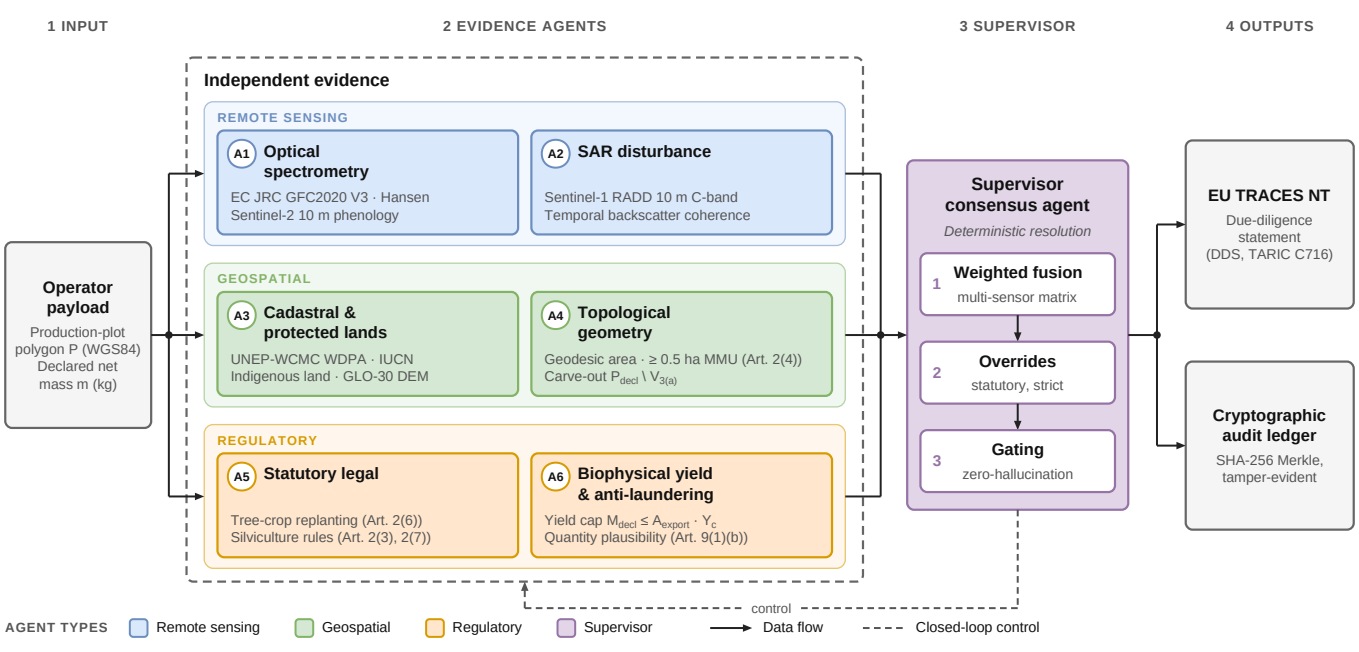
1 INPUT
2 EVIDENCE AGENTS
3 SUPERVISOR
4 OUTPUTS
Independent evidence
REMOTE SENSING
A1 Optical spectrometry
EC JRC GFC2020 V3 · Hansen
Sentinel-2 10 m phenology
A2 SAR disturbance
Sentinel-1 RADD 10 m C-band
Temporal backscatter coherence
GEOSPATIAL
A3 Cadastral & protected lands
UNEP-WCMC WDPA · IUCN
Indigenous land · GLO-30 DEM
A4 Topological geometry
Geodesic area · ≥ 0.5 ha MMU (Art. 2(4))
Carve-out $P_{decl} \setminus V_{3(a)}$
REGULATORY
A5 Statutory legal
Tree-crop replanting (Art. 2(6))
Silviculture rules (Art. 2(3), 2(7))
A6 Biophysical yield & anti-laundering
Yield cap $M_{decl} \le A_{export} \cdot Y_c$
Quantity plausibility (Art. 9(1)(b))
Operator payload
Production-plot polygon P (WGS84)
Declared net mass m (kg)
Supervisor consensus agent
Deterministic resolution
1 Weighted fusion
multi-sensor matrix
2 Overrides
statutory, strict
3 Gating
zero-hallucination
EU TRACES NT
Due-diligence statement (DDS, TARIC C716)
Cryptographic audit ledger
SHA-256 Merkle, tamper-evident
control
AGENT TYPES
Remote sensing
Geospatial
Regulatory
Supervisor
Data flow
Closed-loop control